The equation of the ellipse shown in Fig. 20.39 is

$$\frac{y^2}{\mu_o^2}+\frac{z^2}{\mu_e^2}=1 \qquad (20.23)$$

Substituting equ. (20.31) and equ. (20.32) into equ. (20.33), we obtain

$$\frac{1}{\mu_e^2(\theta)}=\frac{\cos^2\theta}{\mu_o^2}+\frac{\sin^2\theta}{\mu_e^2} \qquad (20.34)$$

Thus, for $\theta = 0^o$, that is propagation along the optic axis, $\mu_e(0°) = n_0$ while for $\theta = 90^o$, $\mu_e(90°) = \mu_e$. The two polarizations which can be propagated correspond to the maximum and minimum refractive indices given by the index ellipsoid. For propagation parallel to the optic axis (that is z-direction), there is no birefringence as the section of the ellipsoid perpendicular to this direction is a circle. For propagation perpendicular to the optic axis, the birefringence will be a maximum, the permitted polarizations will be parallel to the y-axis with refractive index μ_o and parallel to the z-axis with refractive index μ_e.

20.17. PHASE DIFFERENCE BETWEEN e-RAY AND o-RAY

We have seen that natural light incident on the surface of an anisotropic crystal undergoes double refraction and produces two plane polarized waves (Fig.20.40a). Let us consider the particular case of a slice of a calcite crystal (a negative uniaxial crystal) where the optic axis is parallel to refracting face of the crystal. The two waves travel along the same direction in the crystal but with different velocities. As a result, when the waves emerge from the rear face of the crystal, an optical path difference would have developed between them (Fig.20.40b). The optical path difference can be calculated as follows:

Let d be the thickness of the crystal.

$$\left.\begin{array}{l}\text{The optical path for o-ray}\\ \text{within the crystal}\end{array}\right\} = \mu_o d$$

$$\left.\begin{array}{l}\text{The optical path for e-ray}\\ \text{within the crystal}\end{array}\right\} = \mu_e d$$

$$\therefore \quad \left.\begin{array}{l}\text{The optical path difference}\\ \text{between e-ray and o -ray}\end{array}\right\} \Delta = (\mu_o - \mu_e)\, d \qquad \ldots (20.35)$$

Consequently, a phase difference arises between the two waves. It is given by

$$\delta = \frac{2\pi}{\lambda}(\mu_o - \mu_e)\, d \qquad \ldots (20.36)$$

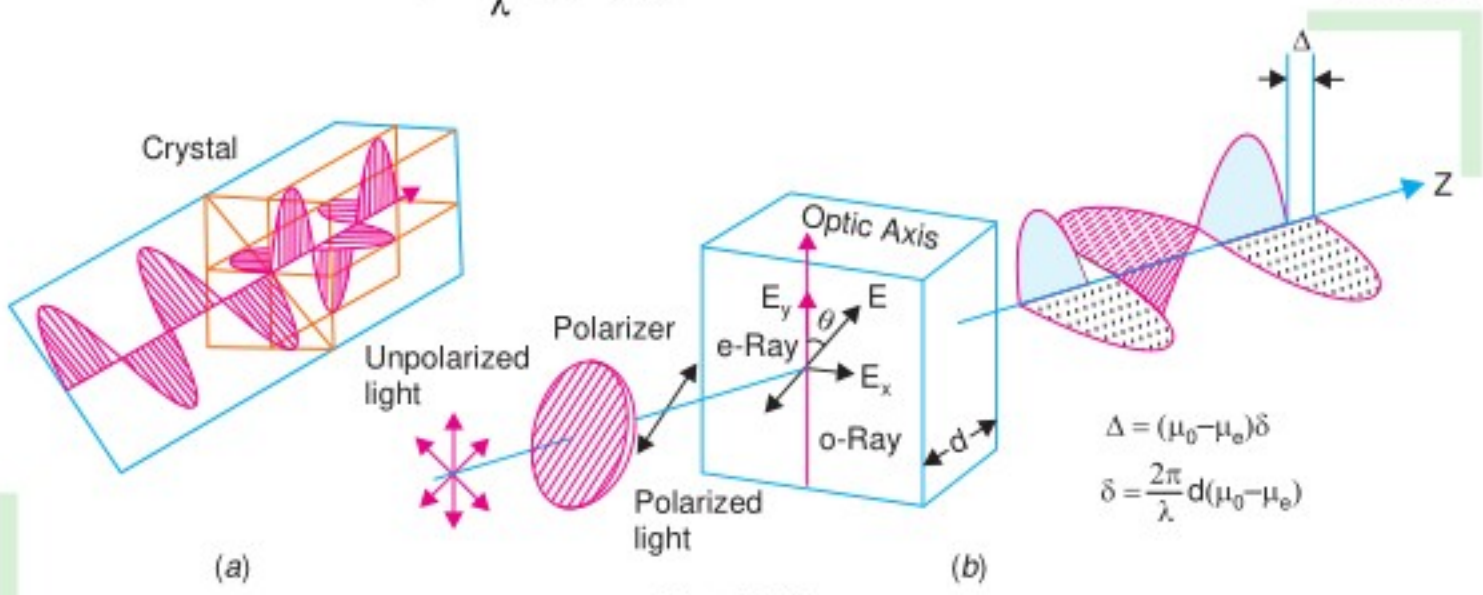


Fig. 20.40

As the two component waves are derived from the same incident wave, the two waves are in phase at the front face and have emerged from the crystal with a constant phase difference and hence it may be expected that the waves are in a position to interfere with each other. However, as the planes of polarization of o-ray and e-ray are perpendicular to each other, interference cannot take place between e- and o-rays. The waves instead combine with each other to give elliptically polarized wave.

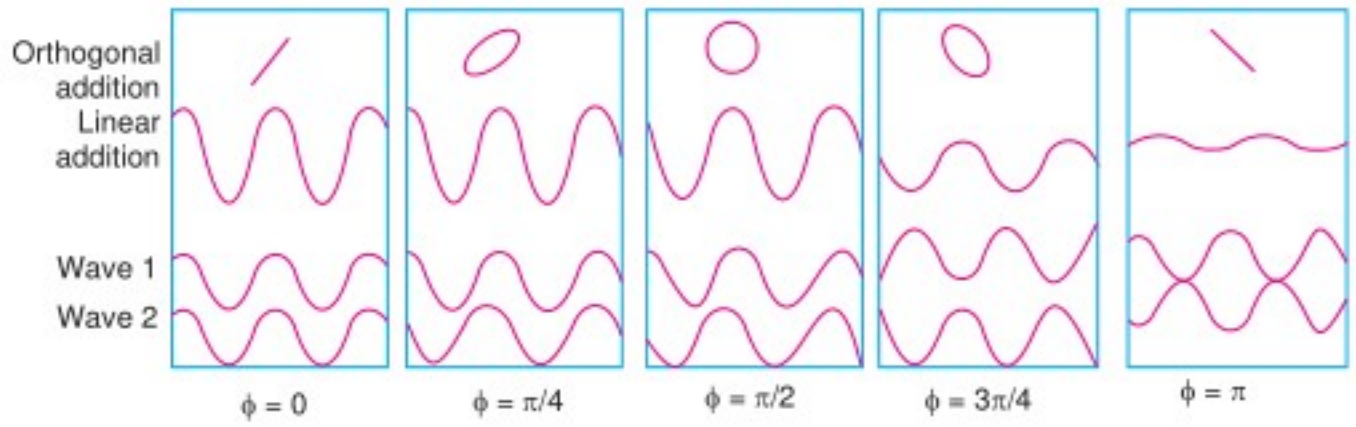


Fig. 20.41

The most important point to be noted here is as follows: The superposition of two coherent waves having a *common plane* of polarization yields a wave polarized in the same plane. The superposition leads to linear addition of waves, i.e., **interference**. The resultant vibration is *linear* and occurs in the same direction as that of the two superposing vibrations (see Fig.20.41). On the other hand, if the two coherent waves are polarized in orthogonal planes, the resultant vibrational motion takes place in two dimensions either in the form of an ellipse, a circle or a straight line depending on the phase difference between the waves (Fig.20.41). This is known as *orthogonal addition.*

20.18. SUPERPOSITION OF WAVES LINEARLY POLARISED AT RIGHT ANGLES

Let us now look at the result of superposition of two waves linearly polarized at right angles to each other (see Fig.20.42 a). Let us consider two light waves travelling in the same direction, x ; one wave is polarized in the xy-plane and the other is polarized in yz-plane. We are interested to know the state of polarization of the resultant wave at the plane x = *constant*.

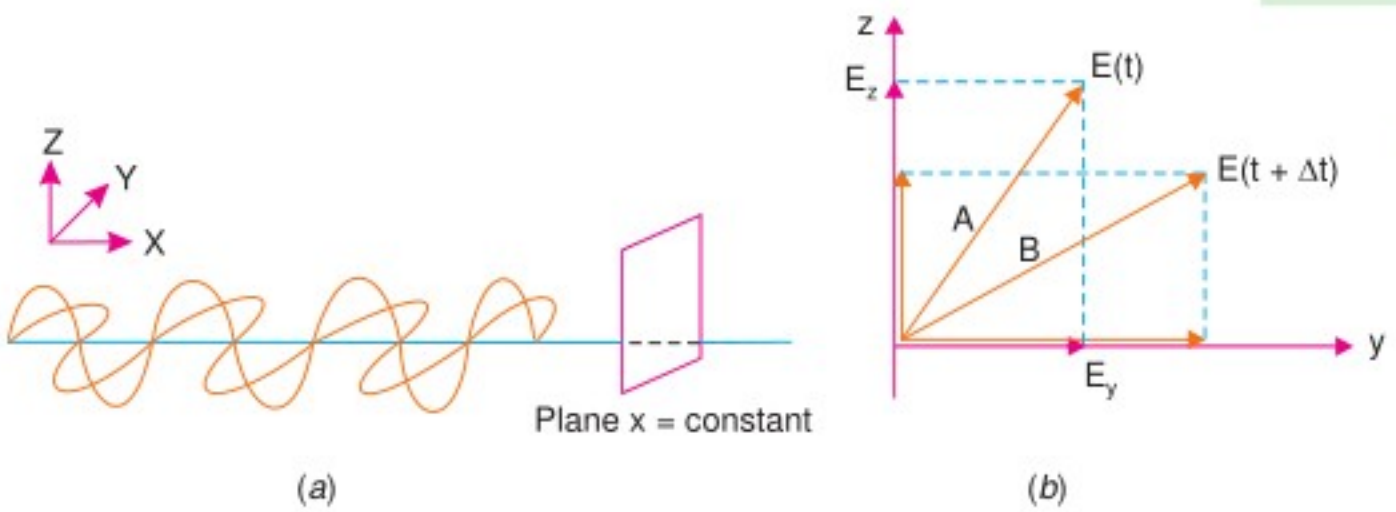


Fig. 20.42

Let the two orthogonal waves be represented by

$$E_y = E_1 \cos\left(k\,x - \omega\,t\right) \tag{20.37}$$

$$E_z = E_2 \cos\left(k x - \omega t + \delta\right) \qquad \ldots (20.38)$$

The waves are of the same frequency $\nu = \omega/2\pi$. δ is the phase difference between the waves. At a given time t, the optical vectors E_y and E_z produce a resultant optical vector of magnitude, say A and at a slightly later time, $t + \Delta t$, they give rise to a resultant vector of magnitude B which points in a different direction (see Fig.20.42b). With progress of time the tip of the resultant optical vector moves along a curve in the yz-plane. We apply the *principle of superposition* to find the equation of the curve traced by the tip of the resultant vector of the two vectors.

According to the principle of superposition

$$\begin{aligned} E &= E_y + E_z \\ &= E_1 \cos(kx - \omega t) + E_2 \cos(kx - \omega t + \delta) \end{aligned} \qquad \ldots (20.39)$$

The equation of the curve may be found by eliminating 't' from the equations. We can write the expansion of equ. (20.39) as

$$\begin{aligned} E_z &= E_2 \cos(kx - \omega t)\cos\delta - E_2 \sin(kx - \omega t)\sin\delta \\ &= E_2 \cos(kx - \omega t)\cos\delta \pm [1 - \cos^2(kx - \omega t)]^{1/2} E_2 \sin\delta \end{aligned} \qquad \ldots (20.40)$$

We find from equ.(20.37) that

$$\cos(kx - \omega t) = E_y/E_1$$

$$\therefore \quad E_z = E_2 \frac{E_y}{E_1} \cos\delta \pm \sqrt{1 - \frac{E_y^2}{E_1^2}}\; E_2 \sin\delta \qquad \ldots (20.41)$$

Rearranging the terms, we get

$$\left[E_z - \frac{E_2}{E_1} E_y \cos\delta\right] = \pm\sqrt{1 - \left(\frac{E_y}{E_1}\right)^2}\; E_2 \sin\delta$$

On squaring both the sides, we obtain

$$E_z^2 + \frac{E_y^2 E_2^2}{E_1^2}\cos^2\delta - \frac{2E_y E_z E_2}{E_1}\cos\delta = E_2^2 \sin^2\delta - \frac{E_y^2 E_2^2}{E_1^2}\sin^2\delta$$

Rearranging the terms, we get

$$E_z^2 + \frac{E_y^2 E_2^2}{E_1^2}(\cos^2\delta + \sin^2\delta) - \frac{2E_y E_z E_2}{E_1}\cos\delta = E_2^2 \sin^2\delta \qquad \ldots (20.42)$$

$$E_z^2 + \frac{E_y^2 E_2^2}{E_1^2} - \frac{2E_y E_z E_2}{E_1}\cos\delta = E_2^2 \sin^2\delta$$

Dividing both the sides by E^2_2 and rearranging the terms, we obtain

$$\frac{E_y^2}{E_1^2} + \frac{E_z^2}{E_2^2} - \frac{2E_y E_z}{E_1 E_2}\cos\delta = \sin^2\delta \qquad \ldots (20.43)$$

Equation (20.43) is the general equation of an *ellipse*. Hence, the tip of the resultant vector traces an ellipse in the yz-plane. The ellipse is constrained within a rectangle having sides $2E_1$ and $2E_2$. The major axis makes an angle α with the y-axis (see Fig.20.43).

$$\tan 2\alpha = 2E_1 E_2 \cos\delta / \left(E_1^2 - E_2^2\right)$$

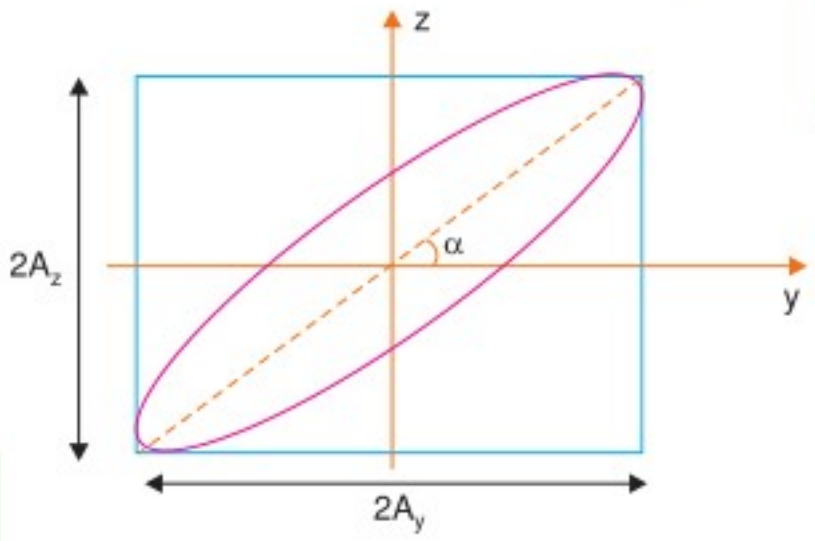


Fig. 20.43

Special cases:

1. When $\delta = 0$, or $\pm 2m\pi$, the two waves are in phase.
   $\cos\delta = 1$ and $\sin\delta = 0$ and the equ. (20.43) reduces to

$$\frac{E_y^2}{E_1^2} + \frac{E_z^2}{E_2^2} - \frac{2E_yE_z}{E_1E_2} = 0$$

$$\left[\frac{E_y}{E_1} - \frac{E_z}{E_2}\right]^2 = 0$$

$$\frac{E_y}{E_1} - \frac{E_z}{E_2} = 0$$

$$\therefore \quad E_z = \frac{E_2}{E_1}E_y \qquad \text{... (20.44)}$$

The above equation represents a straight line, having a slope (E_2/E_1). Therefore, the equation represents a wave having its plane of polarisation making an angle $\tan^{-1}(E_2 - E_1)$ with respect to the y-axis. It means that **the resultant of two plane-polarised waves, which are in phase (*i.e.*, coherent waves), is again a plane-polarised wave.**

2. When $\delta = \pi$, or $\pm (2m+1)\pi$, the two waves are in opposite phase.
   $\cos\delta = -1$ and $\sin\delta = 0$ and the equ. (20.43) reduces to

$$\frac{E_y^2}{E_1^2} + \frac{E_z^2}{E_2^2} - \frac{2E_yE_z}{E_1E_2} = 0$$

$$\left[\frac{E_y}{E_1} + \frac{E_z}{E_2}\right]^2 = 0$$

$$\frac{E_y}{E_1} + \frac{E_z}{E_2} = 0$$

$$\therefore \quad E_z = -\frac{E_2}{E_1}E_y \qquad \text{... (20.45)}$$

This equation represents a straight line of a slope $(-E_2/E_1)$. Therefore, the equation represents a wave having its plane of polarization making an angle $\tan^{-1}(-E_2/E_1)$ with respect to the y-axis. It means that **the resultant of two plane-polarized waves, which are in opposite phase (*i.e.*, coherent waves), is again a plane-polarized wave.**

3. If $\delta = \pi/2$, or $\pm (2m+1)\pi/2$, then $\cos\delta = 0$ and $\sin\delta = 1$. Equ.(20.43) reduces to

$$\frac{E_y^2}{E_1^2} + \frac{E_z^2}{E_2^2} = 1 \qquad \text{... (20.46)}$$

This is the equation of an ellipse whose major axis and minor axis coincide with y- and z-co-ordinate axes. Therefore, when the two plane polarized waves are out of phase by 90°, their resultant is an elliptically polarized wave.

4. In the particular case, when $\delta = \pi/2$ and $E_1 = E_2 = E_o$, equ.(20.43) reduces to

$$E_y^2 + E_z^2 = E_o^2 \qquad \text{... (20.47)}$$

This is the equation of a circle. Hence the resultant light is circularly polarized.

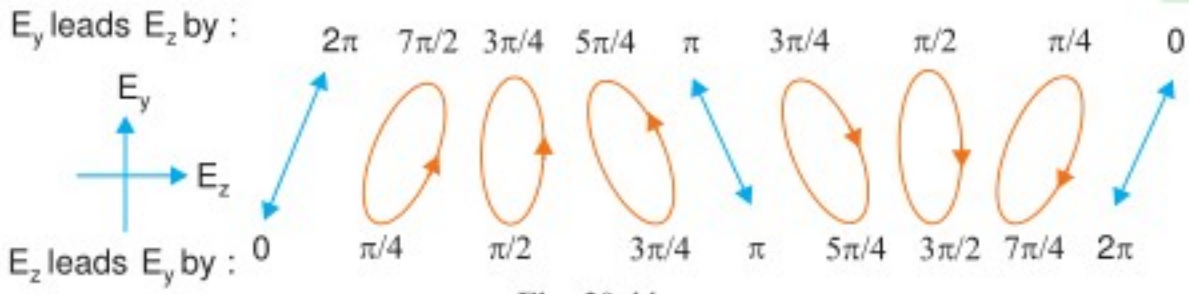


Fig. 20.44

Fig.20.44 shows more generally how the E vector changes with time for various values of δ.

20.18.1 SUPERPOSITION OF e-RAY AND O-RAY:

Now we are in a position to understand what happens when e-ray and o- ray overlap on each other after emerging from an anisotropic crystal plate. It is obvious that they cannot produce interference fringes as in a double slit experiment, since the o-ray are not vibrating in the same plane. On the other hand, they combine to produce different states of polarization depending upon their optical path difference.

1. When the optical path difference is 0 or an even or odd multiple of λ/2, the resultant light wave is linearly polarized.
2. When the optical path difference is λ/4, the resultant light wave is **elliptically polarized.**
3. In the particular instance when the wave amplitudes are equal and the optical path difference is λ/4, the resultant light wave is **circularly polarized**.

20.19. RETARDERS

A **retarder** is a uniform plate of birefringent material whose optic axis lies in the plane of the plate. Retarders are called quarter-wave plates, half-wave plates and full-wave plates depending on their action. They divide the incident wave into two polarized waves that travel perpendicular to the plate at different speeds. A phase retardation of one wave relative to the other is therefore introduced as the waves cross the thickness *d* of the plate. They are used to produce circularly or elliptically polarised light and to analyse polarised light into its elliptical components.

20.19.1 QUARTER WAVE PLATE

A *quarter wave plate* is a thin plate of birefringent crystal having the optic axis parallel to its refracting faces and its thickness adjusted such that it introduces a quarter-wave (λ/4) path difference (or a phase difference of 90°) between the e-ray and o-ray propagating through it.

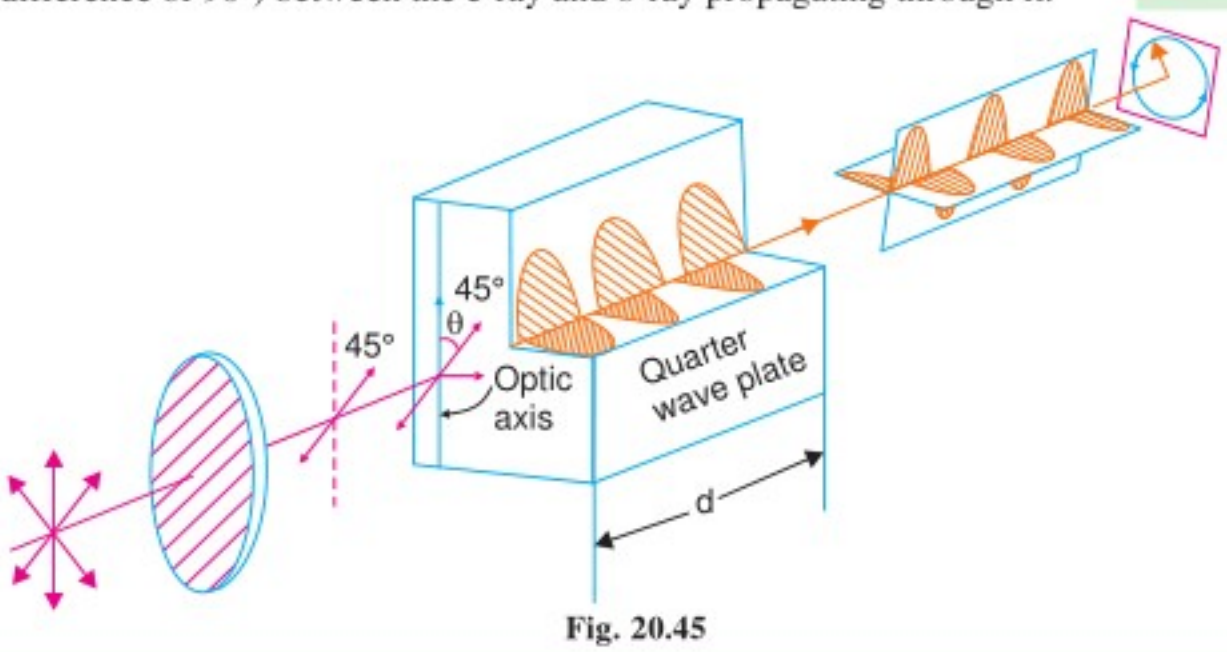


Fig. 20.45

When a plane polarized light wave is incident on a negative birefringent crystal having the optic axis parallel to its refracting face, the wave splits into e-wave and o-wave (Fig.20.45). The two waves travel along the same direction but with different velocities. As a result, when they emerge from the rear face of the crystal, an optical path difference would be developed between them. Thus,

$$\therefore \qquad (\mu_o - \mu_e)d = \frac{\lambda}{4} \qquad \ldots (20.48)$$

$$d = \frac{\lambda}{4(\mu_o - \mu_e)} \qquad \ldots (20.49)$$

A quarter wave plate introduces a phase difference δ, between e-ray and o-ray given by $\delta = (2\pi/\lambda)\Delta = \pi/2 = 90°$.

A quarter-wave plate is used for producing elliptically or circularly polarized light. It converts plane-polarized light into elliptically or circularly polarized light depending upon the angle that the incident light vector makes with the optic axis of the quarter wave plate.

20.19.2 HALF WAVE PLATE

A *half wave plate* is a thin plate of birefringent crystal having the optic axis parallel to its refracting faces and its thickness chosen such that it introduces a half-wave ($\lambda/2$) path difference (or a phase difference of 180°) between e-ray and o-ray.

When a plane polarized light wave is incident on a birefringent crystal having the optic axis parallel to its refracting faces, it splits into two waves: o- and e-waves. The two waves travel along the same direction inside the crystal but with different velocities. As a result, when they emerge from the rear face of the crystal, an optical path difference would be developed between them.

$$\therefore \qquad (\mu_o - \mu_e)d = \frac{\lambda}{2} \qquad \ldots (20.50)$$

$$\therefore \qquad d = \frac{\lambda}{2(\mu_o - \mu_e)} \qquad \ldots (20.51)$$

A half wave plate introduces a phase difference δ, between e-ray and o-ray given by $\delta = (2\pi/\lambda)\Delta = \pi = 180°$.

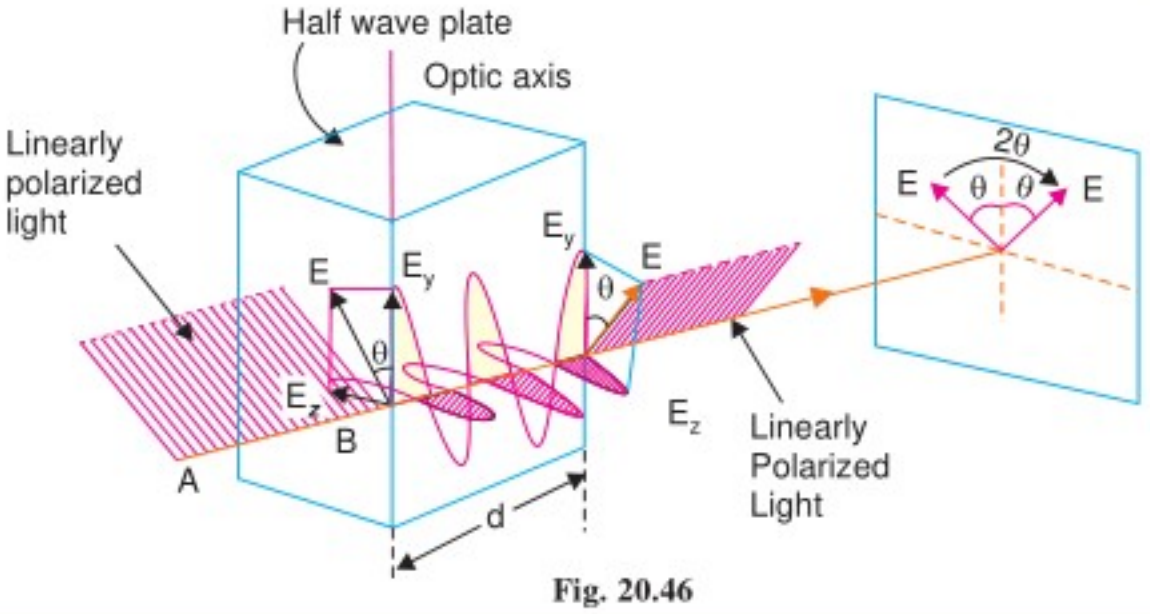


Fig. 20.46

Now let a plane polarized light be incident normally on the half-wave plate. Let the electric vector E make an angle with the optic axis of the half wave plate (See Fig.20.46). The incident wave splits into two waves, e- and o-waves. The waves progressively develop path difference as

they travel through the crystal and they emerge with a phase difference of 180°. When the two waves combine, they yield a plane-polarized wave, which has its plane of polarization rotated through an angle of 2θ.

Therefore, ***a half-wave plate rotates the plane of polarization of the incident plane polarized light through an angle 2θ.*** The half wave plate will invert the handedness of elliptical or circular polarized light, changing right to left and vice versa.

Now we are in a position to understand what happens when e-ray and o- ray overlap on each other after emerging from an anisotropic crystal plate. It is obvious that they cannot produce interference fringes as in a double slit experiment. On the other hand, they combine to produce different states of polarization depending upon their optical path difference.

1. When the optical path difference is 0 or an even or odd multiple of λ/2, the resultant light wave is linearly polarized.
2. When the optical path difference is λ/4, the resultant light wave is **elliptically polarized.**
3. In the particular instance when the wave amplitudes are equal and the optical path difference is λ/4, the resultant light wave is **circularly polarized**.

20.20. PRODUCTION OF ELLIPTICALLY POLARIZED LIGHT

A quarter wave plate and a polarizer are the optical devices necessary to produce elliptically polarized light from unpolarized light.

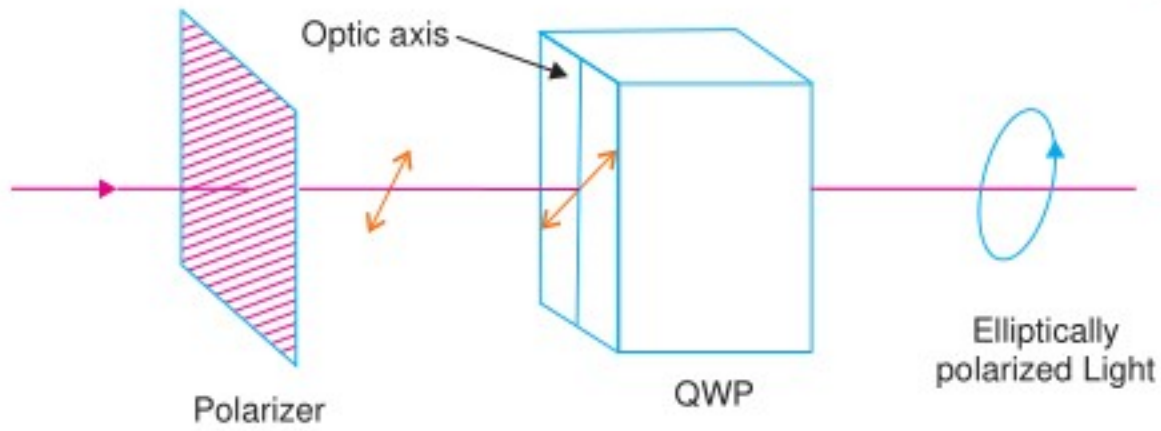


Fig. 20.47

Unpolarized light is first converted to plane polarized light by allowing it to pass through a polarizer (a polaroid sheet or a Nicol prism), as shown in Fig.20.47. The plane-polarized light is then made incident on a quarter wave plate. The quarter wave plate or the polarizer is rotated such that the electric vector E of plane polarized light wave makes an angle θ (≠45°) with the optic axis of the quarter wave plate. The incident ray divides into o-ray and e-ray of amplitudes E sin θ and E cos θ. The rays travel along the same direction in the crystal with different velocities. The two rays are polarized in orthogonal planes. They are in phase at the front face but progressively get out of phase as they travel through the crystal. When they emerge out of the crystal they will have a path difference of λ /4 or a phase difference of 90°. When they combine, they produce elliptically polarized light.

20.20.1 DETECTION OF ELLIPTICALLY POLARIZED LIGHT

The light beam is allowed to pass through an analyzer (a polaroid sheet or a Nicol prism). If on rotating the analyzing Polaroid sheet or Nicol, the intensity of the emerging beam varies from a maximum to a minimum value, but never reaching zero, then the incident light is elliptically polarized.